

SWOT

✔ Strengths (Internal – What we do well)

- User-friendly interface – Designed for easy navigation for students, teachers, and admins.
- Multi-role dashboards – Tailored views and functionalities for students, teachers, and admins.
- Comprehensive LMS features – Includes courses, assignments, quizzes, grades, and progress tracking.
- Modular content delivery – Organized into structured modules for focused learning.
- Integrated AI chatbot – Provides instant student support and handles FAQs.
- Responsive design – Accessible on both desktop and mobile browsers.
- Automated notifications – Sends email and SMS reminders for assignments, grades, and deadlines.
- Modern integrations – Email (PHPMailer), SMS, AI chatbot, and payment systems integrated.
- Payment and enrollment tracking – Built-in transaction and enrollment management.
- User engagement tools – Ratings, reviews, messaging, and notifications improve interaction.
- Content variety – Supports videos, PDFs, documents, and various file types.
- Progress tracking and analytics – Students monitor progress; teachers/admins get performance insights.
- E-certificates – Rewards learners and motivates course completion.
- Monetization ready – Payment processing for course fees and premium modules.

✖ Weaknesses (Internal – What needs improvement)

- Limited real-time collaboration – No live video sessions or chat rooms initially.
- No dedicated mobile app – Web-only access limits mobile usability.
- Manual grading – Some assignments require manual checking.

- Scalability concerns – Initial design may struggle with large user volumes.
 - Dependency on third-party APIs – Email, SMS, payment gateways, and AI services.
 - Security and data protection – Requires strong implementation for authentication and transactions.
 - Complex integrations – Multiple external services increase technical complexity.
 - Testing limitations – Limited time may restrict thorough QA of all features.
 - Team experience – Developers may need to learn new tools or APIs (PHP, integrations, AI).
 - Time and resource constraints – Tight timeline (e.g., 11 weeks) and potential budget limits for paid APIs.
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Opportunities (External – Market/trends we can use)

- Growing demand for online education – Ongoing post-pandemic shift to e-learning.
 - Remote and mobile learning trend – Supports the future development of a companion mobile app.
 - Microlearning trend – Can expand into short-form or skill-based courses.
 - Corporate training potential – Adaptable for company-based learning platforms.
 - Gamification – Badges, leaderboards, or certificates to increase engagement.
 - Integration with popular tools – Future links with Google Classroom, Zoom, or Microsoft Teams.
 - AI-driven personalization – Recommend courses based on user behavior.
 - Niche targeting – Specialize in certain subjects or audiences.
 - Partnership opportunities – Collaborate with schools, institutions, or content creators.
 - Blockchain-verified certifications – Enhance e-certificate integrity in the future.
 - Global reach – Internet-based access opens to international learners.
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Threats (External – Risks and challenges)

- Strong competition – Platforms like Moodle, Canvas, Udemy, Coursera, and Blackboard dominate the market.
- Changing tech standards – Must keep up with security, UI, and accessibility requirements.
- Data privacy compliance – Follow GDPR and local data protection laws.
- API dependency risks – Changes in pricing, functionality, or service downtime can disrupt operations.
- User adoption – Teachers or students may prefer familiar systems.
- Technical challenges – Risks of bugs, integration failures, and system downtime.
- Funding and maintenance – Ongoing server, domain, and API costs.
- Scope creep – Feature lists may expand beyond the timeline or capacity.
- Content moderation – Ensuring course quality and authenticity.
- Payment management – Handling refunds, disputes, or fraudulent transactions.
- Technical skill gaps – Upgrading features (AI, real-time tools) may exceed current expertise.

CHEN

1. USERS

text

userID (PK) ← Primary Key (underline)

firstName (required)

lastName (required)

middleInitial (optional) ← dashed oval

email (required, unique)

passwordHash (required)

profilePictureURL (optional)

phoneNumber (optional)

lastLogin (optional)

createdAt (required)

updatedAt (optional)

userStatusID (FK → USER_STATUS) ← line to USER_STATUS

2. ROLES

text

roleID (PK)

roleName (required) ← values: student, teacher, admin, teaching_assistant

description (optional)

3. USER_ROLES

text

userRoleID (PK)

userID (FK → USERS) ← line to USERS

roleID (FK → ROLES) ← line to ROLES

assignedAt (required)

isActive (required)

4. USER_STATUS

text

userStatusID (PK)

statusName (required) ← values: active, inactive, suspended, deleted

description (optional)

5. CATEGORIES

text

categoryID (PK)

categoryName (required)

description (optional)

parentCategoryID (FK → CATEGORIES) ← optional, line to CATEGORIES (self-reference)

createdAt (required)

6. COURSE_STATUS

text

courseStatusID (PK)

statusName (required) ← values: draft, published, archived, under_review

description (optional)

7. COURSES

text

courseID (PK)

categoryID (FK → CATEGORIES) ← line to CATEGORIES

instructorID (FK → USERS) ← line to USERS

courseStatusID (FK → COURSE_STATUS) ← line to COURSE_STATUS

title (required)

description (optional)
price (required)
thumbnailURL (optional)
difficultyLevel (required) ← values: beginner, intermediate, advanced
estimatedHours (optional)
totalEnrollments (optional) ← dashed oval, for caching
avgRating (optional) ← dashed oval, for caching (1-5 scale)
createdAt (required)
updatedAt (optional)

8. MODULES

text
moduleID (PK)
courseID (FK → COURSES) ← line to COURSES
title (required)
description (optional)
orderIndex (required)
createdAt (required)

9. CONTENT

text
contentID (PK)
moduleID (FK → MODULES) ← line to MODULES
contentType (required) ← values: video, pdf, document, text, link ← FIXED: removed quiz, assignment
title (required)
description (optional)
contentURL (optional)

durationMinutes (optional) ← dashed oval

orderIndex (required)

createdAt (required)

10. ENROLLMENT_STATUS

text

enrollmentStatusID (PK)

statusName (required) ← values: active, completed, dropped, expired

description (optional)

11. ENROLLMENTS

text

enrollmentID (PK)

userID (FK → USERS) ← line to USERS

courseID (FK → COURSES) ← line to COURSES

enrollmentStatusID (FK → ENROLLMENT_STATUS) ← line to ENROLLMENT_STATUS

enrolledAt (required)

completedAt (optional) ← dashed oval

lastAccessed (optional) ← dashed oval

12. PROGRESS

text

progressID (PK)

enrollmentID (FK → ENROLLMENTS) ← line to ENROLLMENTS

contentID (FK → CONTENT) ← line to CONTENT

isCompleted (required)

completedAt (optional) ← dashed oval

timeSpentMinutes (optional) ← dashed oval

lastAccessed (optional) ← dashed oval

13. CERTIFICATES

text

certificateID (PK)

userID (FK → USERS) ← line to USERS

courseID (FK → COURSES) ← line to COURSES

certificateNumber (required, unique)

issuedAt (required)

expiryDate (optional) ← dashed oval

verificationCode (required, unique)

downloadURL (optional) ← dashed oval

14. ASSIGNMENTS

text

assignmentID (PK)

moduleID (FK → MODULES) ← line to MODULES

title (required)

description (optional)

instructions (optional) ← dashed oval

dueDate (required)

maxPoints (required)

submissionType (required) ← values: file_upload, text_entry, both

createdAt (required)

15. SUBMISSIONS

text

submissionID (PK)

assignmentID (FK → ASSIGNMENTS) ← line to ASSIGNMENTS

userID (FK → USERS) ← line to USERS

submissionURL (optional) ← dashed oval

comments (optional) ← dashed oval

submittedAt (required)

gradedAt (optional) ← dashed oval

16. GRADES

text

gradeID (PK)

submissionID (FK → SUBMISSIONS) ← line to SUBMISSIONS

graderID (FK → USERS) ← line to USERS

score (required)

feedback (optional) ← dashed oval

gradedAt (required)

17. QUIZZES

text

quizID (PK)

moduleID (FK → MODULES) ← line to MODULES

title (required)

description (optional)

passingScore (required)

timeLimitMinutes (optional) ← dashed oval

maxAttempts (optional) ← dashed oval

shuffleQuestions (required)

showResults (required)

createdAt (required)

18. QUIZ_QUESTIONS

text

questionID (PK)

quizID (FK → QUIZZES) ← line to QUIZZES

questionType (required) ← values: multiple_choice, true_false

questionText (required)

points (required)

orderIndex (required)

19. QUESTION_OPTIONS

text

optionID (PK)

questionID (FK → QUIZ_QUESTIONS) ← line to QUIZ_QUESTIONS

optionText (required)

optionLetter (required) ← values: A, B, C, D, T, F

isCorrect (required)

orderIndex (optional) ← dashed oval

20. QUIZ_ATTEMPTS

text

attemptID (PK)

quizID (FK → QUIZZES) ← line to QUIZZES

userID (FK → USERS) ← line to USERS

score (required)

timeSpentMinutes (optional) ← dashed oval

startedAt (required)

completedAt (required)

21. ATTEMPT_ANSWERS

text

answerID (PK)

attemptID (FK → QUIZ_ATTEMPTS) ← line to QUIZ_ATTEMPTS

questionID (FK → QUIZ_QUESTIONS) ← line to QUIZ_QUESTIONS

selectedOptionID (FK → QUESTION_OPTIONS) ← optional, dashed line to QUESTION_OPTIONS

textAnswer (optional) ← dashed oval

isCorrect (required)

pointsEarned (required)

22. MESSAGES

text

messageID (PK)

senderID (FK → USERS) ← line to USERS

receiverID (FK → USERS) ← line to USERS

subject (optional) ← dashed oval

messageText (required)

isRead (required)

sentAt (required)

23. ANNOUNCEMENTS

text

announcementID (PK)

courseID (FK → COURSES) ← optional, dashed line to COURSES

authorID (FK → USERS) ← line to USERS

announcementType (required) ← values: system_wide, course, urgent, maintenance

title (required)

content (required)

priority (required) ← values: low, medium, high, urgent

postedAt (required)

expiresAt (optional) ← dashed oval

24. NOTIFICATIONS

text

notificationID (PK)

userID (FK → USERS) ← line to USERS

notificationType (required) ← values: system, course, message, grade, payment

title (required)

message (required)

relatedEntityType (required) ← values: course, assignment, quiz, payment, user, module, content

relatedEntityID (required)

isRead (required)

createdAt (required)

25. REVIEWS

text

reviewID (PK)

userID (FK → USERS) ← line to USERS

courseID (FK → COURSES) ← line to COURSES

rating (required) ← values: 1-5

title (optional) ← dashed oval

comment (optional) ← dashed oval

createdAt (required)

26. ACTIVITY_LOGS

text

logID (PK)

userID (FK → USERS) ← line to USERS

activityType (required) ← values: login, logout, course_view, content_view, quiz_taken, assignment_submitted, payment_made, chatbot_used

action (required)

details (optional) ← dashed oval

ipAddress (optional) ← dashed oval

userAgent (optional) ← dashed oval

createdAt (required)

27. PAYMENT_METHODS

text

methodID (PK)

methodName (required) ← values: credit_card, paypal, bank_transfer, cryptocurrency

provider (optional) ← dashed oval

isActive (required)

28. PAYMENT_STATUS

text

paymentStatusID (PK)

statusName (required) ← values: pending, completed, failed, refunded, disputed

description (optional) ← dashed oval

29. PAYMENTS

text

paymentID (PK)

userID (FK → USERS) ← line to USERS

courseID (FK → COURSES) ← line to COURSES

methodID (FK → PAYMENT_METHODS) ← line to PAYMENT_METHODS

paymentStatusID (FK → PAYMENT_STATUS) ← line to PAYMENT_STATUS

amount (required)

currency (required)

transactionID (optional, unique) ← dashed oval

paymentDate (required)

30. PAYOUTS

text

payoutID (PK)

instructorID (FK → USERS) ← line to USERS

paymentStatusID (FK → PAYMENT_STATUS) ← line to PAYMENT_STATUS

amount (required)

currency (required)

commissionRate (required)

requestDate (required)

processedDate (optional) ← dashed oval

transactionID (optional, unique) ← dashed oval

31. DISPUTE_REASONS

text

reasonID (PK)

reasonName (required) ← values: payment_not_received, unauthorized_charge,
service_not_provided, quality_issue

description (optional) ← dashed oval

32. DISPUTES

text

disputeID (PK)

paymentID (FK → PAYMENTS) ← line to PAYMENTS

reportedByID (FK → USERS) ← line to USERS

reasonID (FK → DISPUTE_REASONS) ← line to DISPUTE_REASONS

description (required)

paymentStatusID (FK → PAYMENT_STATUS) ← line to PAYMENT_STATUS

resolution (optional) ← dashed oval

resolvedByID (FK → USERS) ← optional, dashed line to USERS

createdAt (required)

resolvedAt (optional) ← dashed oval

NEW NEW TABLES FOR AI CHATBOT:

33. CHATBOT_CONVERSATIONS (NEW)

text

chatbotConversationID (PK)

userID (FK → USERS) ← line to USERS

sessionID (required, unique)

startedAt (required)

endedAt (optional) ← dashed oval

isActive (required)

34. CHATBOT_MESSAGES (NEW)

text

chatbotMessageID (PK)

conversationID (FK → CHATBOT_CONVERSATIONS) ← line to CHATBOT_CONVERSATIONS

senderType (required) ← values: user, bot

messageText (required)

sentAt (required)